

Code:

```
#include <stdio.h>

#define MAXN 200005
#define LOGN 20

int up[LOGN][MAXN];

void link(int i, int j) {
    up[0][i] = j;
}

int main() {
    int n, q;
    if (scanf("%d %d", &n, &q) != 2) return 0;

    up[0][1] = 0;
    for (int i = 2; i <= n; i++) {
        int boss;
        scanf("%d", &boss);
        link(i, boss);
    }

    // Binary lifting table precomputation
    for (int d = 1; d < LOGN; d++) {
        for (int i = 1; i <= n; i++) {
            int p = up[d - 1][i];
            up[d][i] = (p == 0) ? 0 : up[d - 1][p];
        }
    }

    while (q--) {
        int x, k;
        scanf("%d %d", &x, &k);
        for (int d = 0; d < LOGN; d++) {
            if ((k >> d) & 1) {
                x = up[d][x];
                if (x == 0) break;
            }
        }
        printf("%d\n", x == 0 ? -1 : x);
    }
}
```

```
}  
    return 0;  
}
```

Input:

```
5 3  
1 3 2 2  
4 2  
1 2  
1 3
```

Output:

```
1  
-1  
-1
```